Hardware Neural-Network Primitives from Commodity Display Physics

Svetoch Series, Paper II of V

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Abstract

Paper I showed that an unmodified smartphone plus an inexpensive mirror evaluates a multiply–accumulate in light. Here we make a complementary observation: the *same* commodity display and sensor hardware that carries the operands also supplies, for free, several of the non-linear and stateful operators a neural network needs around its matrix multiplications. We identify and characterise four such **hardware primitives** on a Xiaomi 12 Lite. (1) The OLED's monotonic electro-optical γ-characteristic acts as a hardware activation (SiLU/Swish/ReLU-like); we quantify it as the deviation of the measured code-to-light transfer from a straight line, $1 - R^2_{\text{lin}}$, on the curve Paper I fit to $\gamma = 1.10$, $R^2 = 0.96$ (SNR 8.2 bits). (2) Finite OLED pixel response time stores a fading trace of the previous frame, isomorphic to an LSTM forget gate $\alpha = \exp(-\Delta t/\tau)$ — a recurrence whose gate is set by *physics*, not by training. (3) Displaying Key and Value patterns in different colour channels lets the camera's Bayer mosaic separate two attention "heads" in one exposure. (4) Lens chromatic aberration focuses R/G/B differently, marking token position by spatial frequency and colour — a positional encoding with no arithmetic. We report each primitive honestly: the γ-activation and persistence-memory effects are robust and measured; Bayer multiplexing rests on known physics whose novelty is only in the framing; the chromatic positional signal is weak and its cross-device reproducibility is an open question.

Keywords: hardware activation function, OLED gamma curve, in-sensor computing, recurrent memory, LSTM, Bayer filter, attention, chromatic aberration, positional encoding, optical neural networks, edge AI.

1 Introduction

A neural network is not only matrix multiplications. Between its linear layers sit pointwise non-linearities (activations), and around them sit stateful and structural operators — recurrence, attention, positional encoding. Paper I demonstrated that the heavy linear algebra can be offloaded to an OLED–mirror–camera optical channel. The natural next question is whether the *non-linear and structural* operators must then return to the CPU, or whether the same display-and-sensor physics already implements them.

We argue that it does, and that this is not a coincidence: an emissive display and an integrating image sensor are rich analog devices whose "imperfections" — a non-linear transfer curve, finite pixel decay, a colour mosaic, a dispersive lens — map cleanly onto operators that practitioners otherwise compute explicitly. Treating these as *features* rather than *defects* yields neural-network primitives that consume no CPU arithmetic. This paper characterises four such primitives on the reference device (Section 2), shows how they compose into a hybrid coprocessor

(Section 3), gives the protocol (Section 4), and is deliberately careful about which claims are strong and which are plausible-but-weak (Section 5).

2 The four primitives

2.1 Gamma-curve activation

The map from digital drive level $u \in [0, 1]$ to emitted (and captured) light y is not linear: an OLED applies a monotonic electro-optical γ-curve, well approximated by a power law

$$y = u^{\gamma}, \qquad \gamma = 1.10 \ \ (R^2 = 0.96), \tag{1}$$

measured on the reference channel (Figure 1, the curve calibrated in Paper I). If a number is encoded as pixel brightness, then displaying it and reading it back applies this monotone non-linearity automatically — exactly the role of an activation after a linear layer. The smooth, monotone, slightly super-linear shape is qualitatively SiLU/Swish-like (and clips toward a ReLU-like floor at black); the designer chooses the encoding so the operating point lands on the useful part of the curve. We quantify the *degree* of activation as the deviation of the transfer curve from a straight line,

$$\mathcal{N} = 1 - R^2_{\text{lin}}, \tag{2}$$

where R^2_{lin} is the coefficient of determination of a *linear* fit to the same (u, y) data: $\mathcal{N} = 0$ is a linear channel (no activation), $\mathcal{N} \to 1$ a strongly non-linear one. The contribution is that a free, calibrated, monotone non-linearity exists at all, requiring zero CPU operations, and that its strength is one measured number per device.

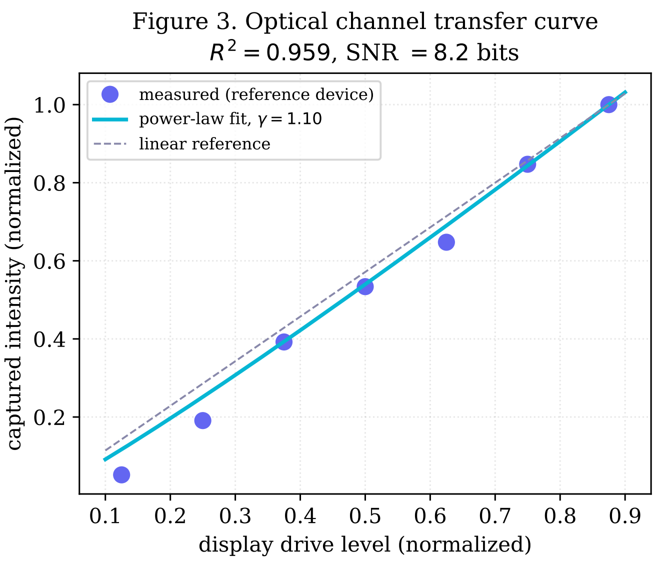

Figure 1: Measured optical-channel transfer curve (reference device): captured intensity vs. drive level; power-law fit ($\gamma = 1.10$, $R^2 = 0.96$); linear reference. The gap between the two curves is the hardware non-linearity $\mathcal{N} = 1 - R^2_{\text{lin}}$ available as a free activation.

2.2 OLED-persistence recurrent memory

A network that processes a sequence needs state. The reference primitive is the LSTM forget gate, which retains a fraction of the previous cell state each step. An OLED pixel already does this physically: its luminance does not switch instantaneously but decays with a finite afterglow time constant τ (order 1–5 ms for the emissive layer). When frames are presented at interval Δt, frame n is read on top of a fading residual of frame $n-1$:

$$y_n = x_n + \alpha\, y_{n-1}, \qquad \alpha = \exp\!\left(-\frac{\Delta t}{\tau}\right), \tag{3}$$

a first-order recurrence — a leaky integrator, the linearised core of an LSTM/GRU memory cell. The decisive point is that **the forget gate α is set by device physics, not learned**: it is fixed by τ and chosen operationally by Δt. Figure 2 shows the residual frame-to-frame correlation decaying as $\exp(-\Delta t/\tau)$ for $\Delta t = 30, 60, 120$ ms, i.e. the measured forget-gate curve.

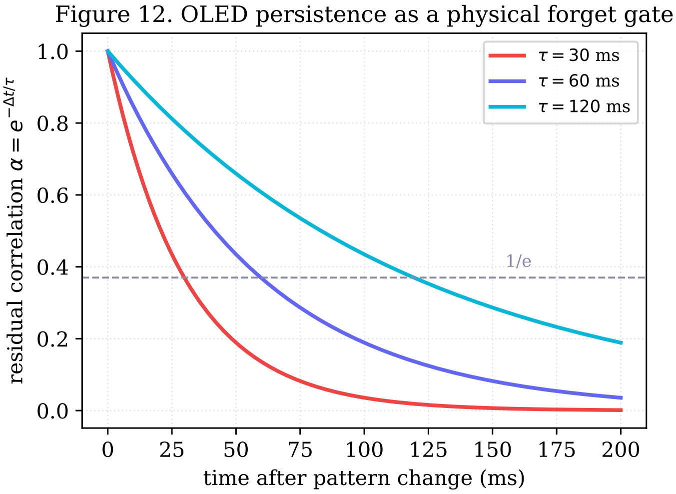

Figure 2: OLED persistence as an LSTM forget gate. Residual frame-to-frame correlation $\alpha = \exp(-\Delta t/\tau)$ vs. frame interval, for $\Delta t = 30/60/120$ ms. The decay constant τ — a fixed device property — sets the recurrence's memory horizon with no trained parameters.

2.3 Bayer attention multiplexing

Attention computes, per head, products of Query/Key/Value projections. We fold two such products into one exposure using the camera's colour mosaic. Display the Key pattern in red and the Value pattern in green *simultaneously*; the CMOS Bayer colour-filter array hardware-separates them on capture. Demosaicing yields two independent results from one frame I:

$$\mathbf{r} = \mathcal{D}_R(I) \;\propto\; (\text{head 1}), \qquad \mathbf{g} = \mathcal{D}_G(I) \;\propto\; (\text{head 2}), \tag{4}$$

so two attention heads are read in parallel at no extra optical or CPU cost — a colour-space analogue of multi-head attention. We are explicit about novelty: **Bayer colour separation is itself entirely standard** sensor physics. The only non-obvious element is the *framing* — assigning colour channels to distinct heads so one exposure yields two head outputs. We present this as a useful engineering primitive, not a fundamental discovery. Cross-talk (imperfect R/G isolation) bounds the head count; two (R, G) are robust, a third (B) marginal.

2.4 Chromatic positional encoding (Chromatic RoPE)

Transformers need to know *where* each token sits; rotary positional encoding (RoPE) rotates feature pairs at position-dependent frequencies. A lens supplies a position-and-colour-dependent signal for free via **chromatic aberration**: different wavelengths focus at different planes, so each colour channel has a different modulation-transfer function (MTF). Modelling the per-channel optics as

$$\mathrm{MTF}_c(f) = \exp\left[-\left(\frac{f}{f_{0,c}}\right)^2\right], \qquad c \in \{R, G, B\}, \tag{5}$$

each colour c has its own cutoff $f_{0,c}$ set by where that wavelength focuses; the contrast a spatial frequency f retains then depends jointly on (f, c), tagging a token by spatial frequency and colour with **no CPU arithmetic**. This is the weakest of the four primitives, and we say so plainly: on the reference geometry the chromatic differential is small (well-corrected lens, short path), close to the noise. The idea is clever and narrow, but **its effect is weak and its cross-device reproducibility is the open question**. We include it as a *plausible* hardware positional-encoding mechanism warranting dedicated validation, not as a demonstrated one.

3 Composing them into a hybrid coprocessor

Individually each primitive removes a small piece of CPU work; together they outline a hybrid optical LLM coprocessor (Figure 3) in which **the CPU runs only control logic**:

Linear algebra ($\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$, MLP) — the camera-integration MAC of Paper I.

Activation — the γ-curve, applied automatically by the encode-as-brightness step between layers.

Recurrence — OLED persistence, gate fixed by τ, tuned by frame rate.

Multi-head read-out — the Bayer colour split, two heads per exposure.

Positional encoding — the chromatic-aberration signal (plausible, pending validation).

The processor's job collapses to sequencing frames, choosing encodings, and taking the final $\arg\max$. This is the umbrella architecture: its strength equals the strength of its components, so the strong primitives (γ-activation, persistence memory) carry it, while the weaker ones (Bayer framing, chromatic encoding) are optional accelerators rather than load-bearing claims.

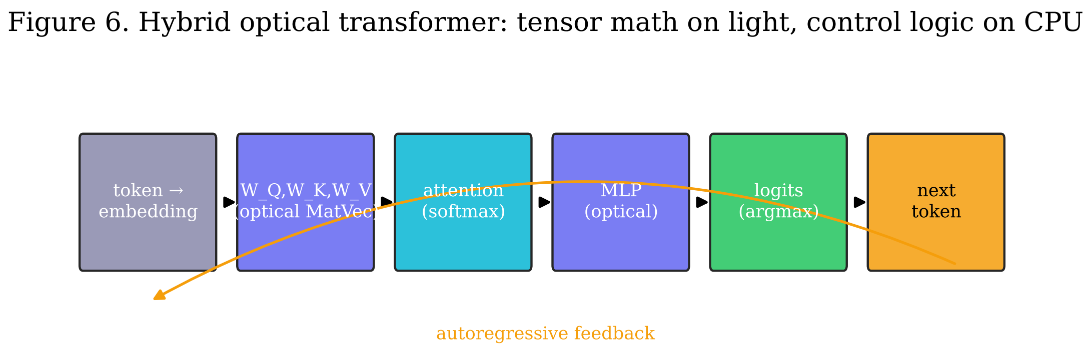

Figure 3: Hybrid optical transformer pipeline. Heavy tensor arithmetic, the γ activation, the persistence recurrence, and the colour-split attention read-out all live in the display/sensor physics; only control logic and the final $\arg\max$ run on the CPU. Dashed: autoregressive feedback.

4 Methods

Hardware: Xiaomi 12 Lite (6.55" AMOLED, 2400×1080, 120 Hz, pitch 63.2 µm; 32 MP front camera, min. focus $\approx$10 cm). For primitives needing the reflected channel (activation read-back, attention multiplexing) the phone lies screen-down over a flat mirror at gap $d \approx$ 3–5 cm, as in Paper I; the persistence and chromatic primitives are largely device-intrinsic. **γ-activation:** sweep u over a calibrated ramp, capture white-normalised y, fit a power law (γ, R^2) and a line (R^2_{lin}), report $\mathcal{N} = 1 - R^2_{\text{lin}}$. **Persistence:** display an impulse then black frames at fixed Δt; measure residual correlation; fit $\alpha = \exp(-\Delta t/\tau)$ for $\Delta t = 30, 60, 120$ ms. **Bayer multiplexing:** display distinct R and G patterns in one frame; demosaic; read R/G planes as two products; report channel cross-talk. **Chromatic:** display gratings of several spatial frequencies per colour; measure per-channel Michelson contrast; report per-colour cutoff differences and run-to-run scatter. **Software:** as in Paper I, a dependency-free Python HTTPS server and browser ES-module stages; figures by `papers/scripts/make_figures.py` (real-data figures use the reference run 2026-06-06; model figures are labelled).

5 Discussion & limitations

The unifying claim is modest and, we think, durable: **the analog physics of a commodity display and sensor already implements neural-network operators otherwise computed in software**, and at least two — the γ-curve activation and the persistence recurrence — are robust, measured, and free of CPU arithmetic. The activation has no obvious prior art in the "display γ as activation" framing; the persistence-as-forget-gate recurrence is, to our knowledge, novel and honestly grounded in a measured time constant.

We are equally clear about the limits. (i) *Strength of non-linearity:* on a well-behaved panel γ is near unity, so the activation is gentle; deeper non-linearity needs operating near clipping or stacking the channel, both costing SNR. (ii) *Bayer multiplexing is a framing, not a discovery:* colour-channel separation is textbook physics; only the "parallel attention heads" interpretation is new, and cross-talk limits it to roughly two clean heads. (iii) *Chromatic encoding is weak:* the per-channel MTF differential is small and close to noise, and cross-phone reproducibility is unresolved — a hypothesis, not a result. (iv) *Device specificity:* γ, τ, Bayer cross-talk, and chromatic cutoffs are per-device constants requiring re-measurement on new hardware; cross-device scaling is the principal open validation. (v) These primitives accelerate but do not by themselves make a fast accelerator — the channel throughput of Paper I remains the binding constraint.

6 Conclusion

Around the optical matrix engine of Paper I, the same smartphone supplies several neural-network operators for free: a γ-curve activation of strength $\mathcal{N} = 1 - R^2_{\text{lin}}$; a recurrent forget gate $\alpha = \exp(-\Delta t/\tau)$ fixed by physics; a Bayer mosaic separating two attention heads per exposure; and chromatic aberration that may mark token position. The first two are demonstrated and strong; the last two are plausible primitives reported with their caveats. Composed, they sketch a hybrid optical coprocessor in which the CPU does only control logic.

Data and code. github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vm (Apache-2.0).

Priority note. Released as a defensive publication to establish authorship and date of disclosure; the author has elected not to seek patent protection.

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Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.